

Oracle-Visibility Selective Injection: A Single-View Reconstruction Diagnostic

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Abstract

Single-view reconstruction is asymmetric: visible regions have image evidence, whereas disoccluded regions require a prior. We study asymmetric latent injection as an *oracle-visibility diagnostic*: the released `visibility.npy` is produced by running VGGT on the complete target clip and therefore crosses the single-image inference boundary. The mask leaves the visible latent path exact, but VAE/decoder spatial mixing prevents a final-RGB guarantee; the direct 16-scene run measured a +0.016 dB visible-PSNR change. A target-GT *quality-gap oracle* correlates with injection benefit at $r = 0.982$ over 166 scenes, while always injecting changes from +2.49 dB on oracle-hard scenes to -2.52 dB on oracle-easy ones. On a fixed 759-frame component diagnostic, the always injected output improves Gen3R PSNR from 17.77 to 19.22 dB and LPIPS-VGG from .360 to .341. Deployment requires a source-only visibility estimator and a source-observable selector; neither is evaluated here.

1 Introduction

Pixels retained from an input view mainly require geometric transport; pixels revealed by camera motion have no direct image evidence and must be generated. Full-image averages obscure this causal distinction. Our central question is: *when does feed-forward geometric evidence improve a generative single-view reconstruction, and how can it be injected without harming already-correct visible regions?*

We treat Flash3D [7] and Gen3R as components rather than leaderboard opponents. Flash3D supplies target-aligned geometric evidence, while the frozen Gen3R backbone supplies a generative prior. A visibility mask routes visible latents around injection exactly. Because geometric evidence is not uniformly better in disocclusions, we analyze optional fallback policies, but the released PSNR-based policies require target GT and are oracle diagnostics rather than deployed model components.

Our three non-overlapping contributions are:

1. Concept: oracle-visibility selective injection as a mechanism diagnostic, not deployable single-view inference.
2. Mechanism: an exact visible latent bypass, with decoded RGB behavior measured rather than guaranteed.
3. Evidence: a scale-stable quality-gap mechanism and perceptual component gains on a fixed long-sequence diagnostic, bounded by failed zero-shot ACID transfer.

2 Related Work

Feed-forward reconstruction. pixelSplat [1], MVSPat [2], and DepthSplat [9] infer Gaussian scenes [4] from two views; Flash3D [7] addresses one view. We use Flash3D as an evidence component, not as an official reproduced baseline.

Generative reconstruction. ViewCrafter [11], CAT3D [3], and GenWarp [6] synthesize unseen content but may drift geometrically. We retain this prior and condition it only where source evidence is absent.

Selective guidance. RePaint [5] conditions known pixels during denoising, while REPA [10] and Geometry Forcing [8] alter training. Our intervention is test-time, region-asymmetric, and allowed to abstain.

3 Method

Let $M \in \{0, 1\}$ denote oracle visibility (1 is visible), z the current Gen3R RGB latent, z_{f3d} encoded evidence, and w a fixed clean-space agreement weight. At each denoising step,

$$z' = M \odot z + (1 - M) \odot [(1 - w)z + wz_{\text{f3d}}]. \quad (1)$$

For $M = 1$, $z' = z$ exactly on the masked latent path. We compute w in clean latent space from

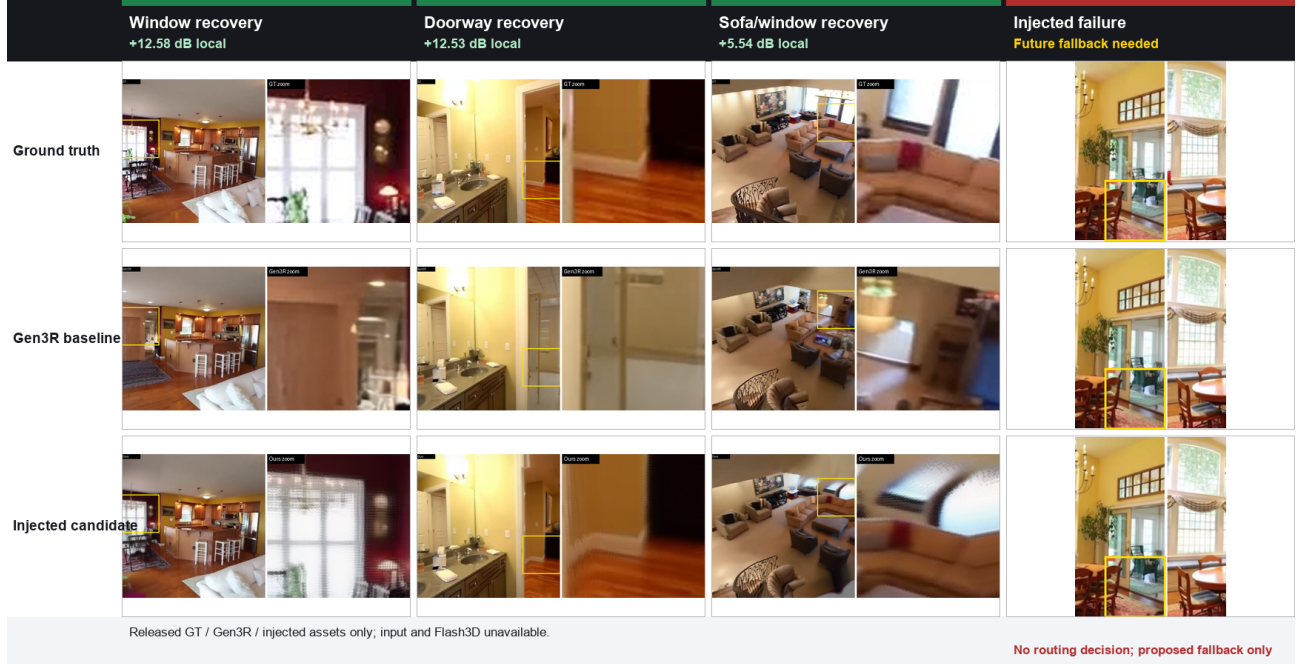


Figure 1: Released qualitative evidence for the fixed long-sequence diagnostic. Columns show three successful recoveries (window, doorway/floor, and sofa/window) and one injection failure. Rows show ground truth, the Gen3R baseline, and the injected candidate; yellow boxes and adjacent zooms use the same released ROI in each row. The first three columns report local ROI PSNR gain. The last column illustrates why a future reliable fallback would be useful; no routing decision is claimed. The teaser is constrained to released GT/Gen3R/injected assets; missing input/Flash3D assets are a release limitation, so the approved design uses the strongest honest teaser available rather than inventing replacements. The figure supports localized recovery and failure analysis, not an official benchmark comparison.

Algorithm 1 Asymmetric geometry-guided generation

- 1: **Input:** image I , target poses P_t , masks M_t
 - 2: $z_{f3d} \leftarrow \text{Encode}(\text{Flash3D}(I, P_t))$
 - 3: $z \leftarrow G(I, P_t)$
 - 4: update only $(1 - M_t)$ during denoising using Eq. (1)
 - 5: **return** $\text{Decode}(z)$
- Optional concept:* a future reliable predictor may return the stored baseline.
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agreement between evidence and the baseline. Because the VAE and decoder mix spatial neighborhoods, the equation does not prove final visible RGB invariance; the direct 16-scene run measured +0.016 dB visible-PSNR change. For analysis only, the *visible-quality proxy* $\text{PSNR}_{f3d, \text{vis}} - \text{PSNR}_{\text{base}, \text{vis}}$ probes relative visible-region quality against target GT; despite using the visible region, it is not available at raw test time. The only released scene-level quantity independent of target GT is VGGT’s `vis_frac`, whose local policy was weak and is not a main result.

4 Experiments

4.1 Protocol and Implementation

Our primary protocol is a **fixed offline oracle-visibility long-sequence diagnostic**, not official MINE/Flash3D evaluation or deployable single-view inference. It uses one designated source image and 48 targets from a 49-frame RealEstate10K [12] sequence. However, VGGT processes the complete target clip to derive `visibility.npy`; all mask/injection results therefore cross the single-image inference boundary. The released main set contains 166 valid scenes. Working renders are 560 pixels; region masks are nearest-neighbor resized. We report masked PSNR and oracle-substitution LPIPS-VGG/SSIM, plus bootstrap 95% CIs from 10,000 scene resamples. Gen3R uses 30 denoising steps and approximately 247 s per scene on one 48 GB GPU. Full protocol declarations and quoted-version metadata are in `../PROTOCOL.md`.

Table 1: Fixed 759-frame full-image component diagnostic on 21 long-sequence scenes selected for significant disocclusion. All rows use one source view, the same stored 560-pixel working renders and metric preprocessing, and LPIPS-VGG/FID/KID full-image metrics; this is neither official MINE nor an official Flash3D reproduction. E-210 evaluates the stored **adaptive2** injected output directly on every included frame, with no oracle fallback. “Evidence ref.” uses per-target scene-folder geometry and is not a single-view leaderboard baseline. Runtime is reported only where released; dashes mean unavailable, not zero. Lower LPIPS/FID/KID is better.

Variant	Views	Frame policy	Resolution	Metric type	PSNR	SSIM	LPIPS	FID	KID	Runtime	Status
Gen3R baseline	1	759 disocc. frames	560 working	full image/VGG	17.77	.653	.360	31.8	.0020	~247 s/scene	reproduced component
Flash3D evidence ref.	1	same frames	560 working	full image/VGG	19.76	.679	.444	54.8	.0096	–	reproduced evidence
Always injected candidate	1	same frames	560 working	full image/VGG	19.22	.676	.341	31.1	.0028	–	injected output

Table 2: Selection analyses on the initial fixed 16-scene diagnostic. Mean and worst are invisible-region PSNR changes in dB; non-selected scenes return the unchanged baseline. All PSNR-based policies use target GT and are oracle probes, including the relative visible-quality proxy; none is a deployment result.

Analysis	Mean	Worst	Injected
Always	+1.58	-6.88	16/16
Oracle invisible	+2.01	-0.92	15/16
Oracle gap > 5	+2.12	-0.17	9/16
Target-GT visible-quality proxy	+2.07	-0.41	14/16
Target-GT absolute F3D	+1.39	-6.88	13/16

4.2 Claim 1: Main Component Diagnostic

The directly evaluated injected output lifts Gen3R PSNR by +1.45 dB and lowers LPIPS-VGG from .360 to .341 while keeping FID close (31.1 vs. 31.8). Flash3D evidence has the highest PSNR but worse LPIPS/FID. These are within-diagnostic component findings, not claims against official Flash3D.

At scene level, Gen3R, Flash3D evidence, always-inject, quality-gap-oracle selectivity, and benefit-oracle selectivity yield invisible PSNR values of 13.78, 19.04, 14.53, 15.44, and 15.49 dB, respectively. Quality-gap-oracle selectivity gains +1.66 dB (95% CI [1.32, 2.02]), wins/ties/loses 85/79/2 scenes at ± 0.05 dB. Its 14.48 dB visible value is assigned from the baseline by the analysis construction, not independently measured. The direct 16-scene RGB run instead measured a +0.016 dB visible change. On 21 high-gain scenes, invisible SSIM changes $.579 \rightarrow .725$ and LPIPS-VGG $.0948 \rightarrow .0784$.

4.3 Claim 2: Oracle Selectivity and Proposed Fallback

On 166 scenes, the target-GT quality-gap oracle at threshold $\tau = 5$ yields +1.658 dB mean and -0.393 dB worst case at a 52% injection rate. Thresholds 3–6 remain within 0.1 dB of the mean-optimal setting. This oracle selectivity quantifies a proposed fallback opportunity, not deployable performance.

4.4 Claim 3: Mechanism, Difficulty, and Scale

The correlation remains .985, .980, .983, .984, and .982 at N=16, 92, 129, 149, and 166; the N=166 bootstrap CI is [.973, .990]. This scale stability supports a mechanism rather than a small-sample association.

4.5 Cross-Dataset Mechanism Transfer

We apply the mechanism without retuning to converted ACID aerial trajectories. This is a diagnostic, not an official leaderboard result. The released E-215 aggregate contains 8 valid scenes: always injection averages -0.443 dB and wins 3/8 scenes. This is failed zero-shot transfer and evidence that selectivity is dataset-dependent, not mechanism success; no per-bucket ACID claim is released.

4.6 Qualitative Successes and Failures

Figure 1 isolates three obvious released recoveries and an injected failure that motivates a future fallback; it does not show a routing decision. Figure 4 provides sequence-level context.

4.7 Published Positioning Only

The exact quoted sources are Flash3D arXiv:2406.04343v2 Table 2; pixelSplat’s camera-ready/retrained README (arXiv:2312.12337v4 Table 1 agrees; checkpoint N/A); arXiv:2403.14627v2 Table 1; checkpoint N/A for MVSplat; and arXiv:2410.13862v3 Table 6; checkpoint N/A for DepthSplat-L. The Flash3D rows are the official +5, +10, and U[-30,30] buckets; full protocol details remain in ../PROTOCOL.md.

5 Limitations

The complete target clip is consumed by VGGT to produce `visibility.npy`; this oracle visibility crosses

Table 3: Quoted published RealEstate10K positioning only; values are not reproduced and no cross-protocol winner is marked. The two-view rows share the official pixelSplat protocol; Flash3D rows share the official MINE protocol. Exact split/index, resolution/crop, target definitions, LPIPS backbone, source/version/table, and checkpoint status are declared in `../PROTOCOL.md`. None uses test-time optimization.

Method	Views	Official target bucket	PSNR	SSIM	LPIPS-VGG
pixelSplat [1]	2	3 interpolated	26.09	.863	.136
MVSplat [2]	2	3 interpolated	26.39	.869	.128
DepthSplat-L [9]	2	3 interpolated	27.47	.889	.114
Flash3D [7]	1	+5	28.46	.899	.100
Flash3D [7]	1	+10	25.94	.857	.133
Flash3D [7]	1	U[-30,30]	24.93	.833	.160

the single-image inference boundary. A deployable system needs a source-only visibility estimator that predicts target visibility from the source image and requested camera alone, plus a source-observable selector. Neither is released or evaluated. The released scene-level policies use target-GT quality probes and are oracle analyses. The latent mask is exact, but final visible RGB invariance is not guaranteed because the VAE/decoder may mix spatially; the direct 16-scene change is +0.016 dB. Large-scale equal visible values are a construction assumption, not an independent measurement. Flash3D evidence is a per-target component reference, not an official single-view reproduction. The teaser is constrained to released GT/Gen3R/injected assets; missing input/Flash3D assets are a release limitation, and this reviewer-visible exception permits the strongest honest teaser rather than invented replacements. Finally, per-frame injection does not optimize temporal consistency: second-order disocclusion energy is .0251 for Gen3R and .0267 for ours over 21 scenes, with ours more stable in 9/21.

6 Conclusion

Under oracle visibility, geometry evidence helps when it is better than the generative prior in newly revealed regions. The exact latent bypass and measured RGB behavior isolate the injection mechanism, while oracle selectivity quantifies a future fallback opportunity. Source-only visibility and selection remain necessary before this diagnostic can become deployable single-view inference.

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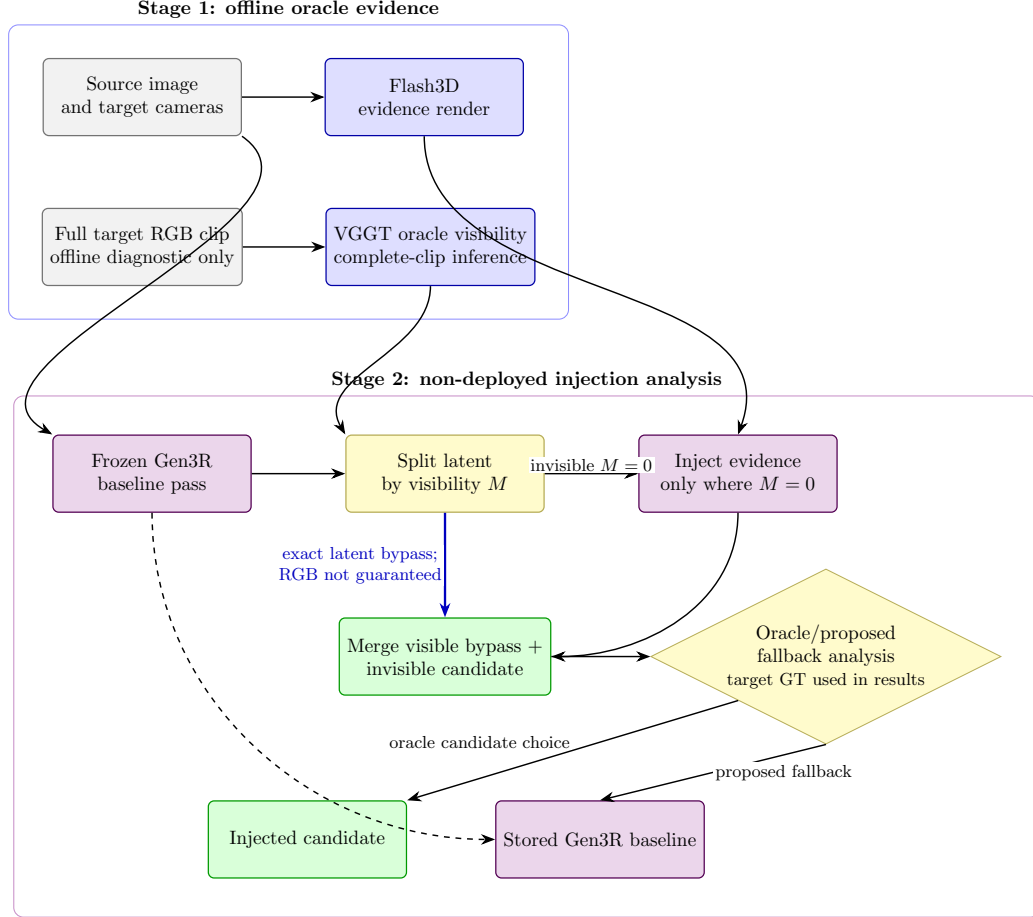


Figure 2: Offline oracle-visibility diagnostic. VGGT processes the complete target clip to construct M , crossing the single-image inference boundary. Stage 2 obtains the frozen Gen3R baseline, preserves the visible latent path exactly, and injects evidence only into invisible latents. This does not guarantee exact visible RGB after VAE decoding. The fallback branch is also oracle analysis using target GT. A deployable method needs source-only visibility and selection estimators; neither is released or evaluated.

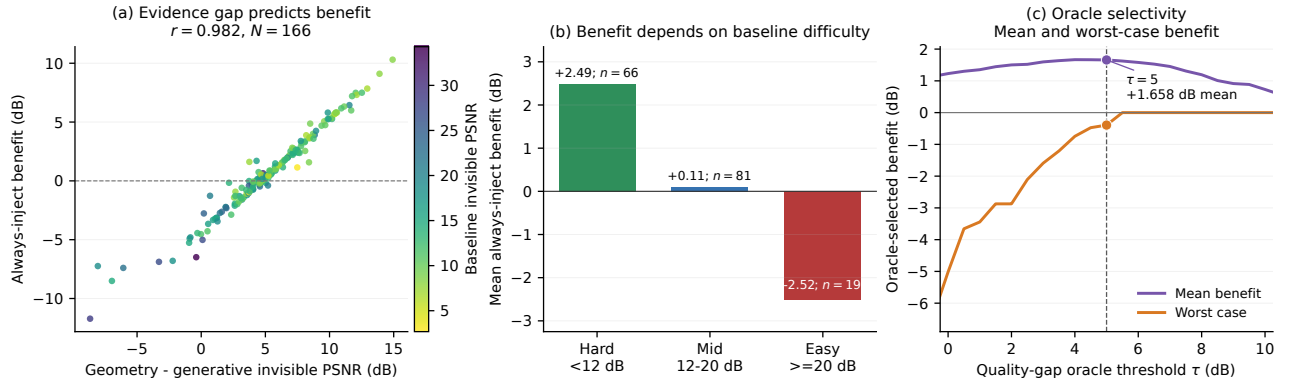


Figure 3: Mechanism evidence derived from released scene JSONs ($N = 166$). (a) Each point is one scene: the Flash3D-evidence minus Gen3R invisible PSNR gap predicts actual always-inject benefit ($r = 0.982$); color denotes baseline difficulty. (b) The same benefit is positive for hard baselines (+2.49 dB), near zero for mid cases (+0.11), and negative for easy cases (−2.52), with bucket counts shown. (c) Sweeping the oracle-gap threshold exposes the mean/tail trade-off; markers show the released default $\tau = 5$ (+1.658 dB mean, −0.393 dB worst). Together the panels show why evidence helps selectively rather than universally.

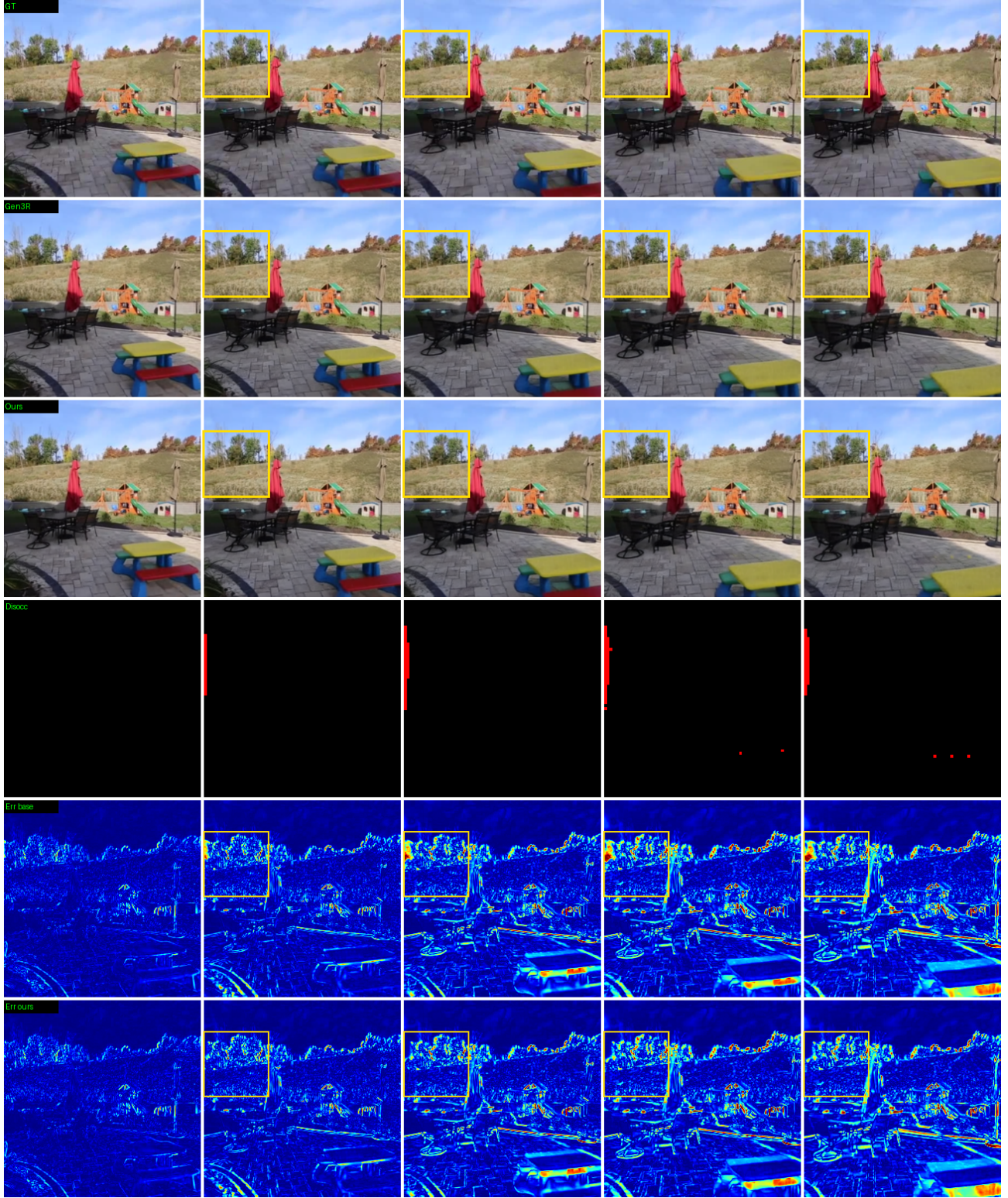


Figure 4: Successful long-sequence component diagnostic. Columns are successive target views. Rows are GT, Gen3R, injected candidate, disocclusion mask, Gen3R error, and candidate error. The same yellow ROI is used across corresponding image and error rows; brighter error denotes larger residual. Error decreases in the newly revealed patio while content outside the ROI is unchanged.

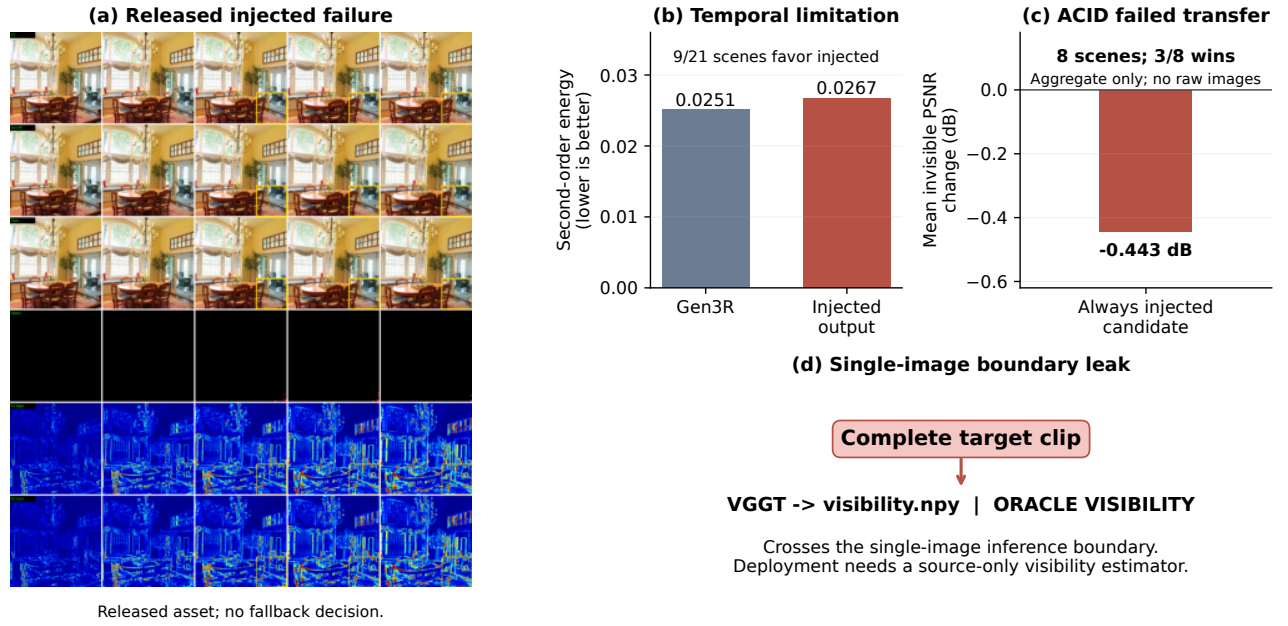


Figure 5: Failure and limitations summary. (a) The released injected candidate fails; no fallback decision exists. (b) Second-order disocclusion energy is .0251 for Gen3R and .0267 for the injected output over 21 scenes, with 9/21 wins. (c) ACID has 8 scenes, a -0.443 dB always-injected mean, and 3/8 wins; no raw ACID images are shown. (d) The critical inference-boundary leak: VGGT consumes the complete target clip to produce `visibility.npy`. Thus all current mask and injection results are offline oracle-visibility diagnostics. Deployment requires a source-only visibility estimator.